

Adarsh Nanjaiya Latha

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Matwenweg 73 - 302, Enschede 7522LD, The Netherlands

RESEARCH PROFILE

I am a Doctoral Candidate in Computer Science at the University of Twente, The Netherlands, specializing in Multimodal AI for drone swarms. My research focuses on integrating Camera, LiDAR and other sensor data to enhance perception in autonomous multi-robot systems. Key contributions include developing information fusion architecture and pipelines that improve collision avoidance, situational awareness in cluttered dynamic environments with applications to search-and-rescue scenarios.

RESEARCH EXPERIENCE

PhD Researcher

University of Twente, The Netherlands

May 2025 - Current

- Conduct research on Multimodal AI and Multi-Agent systems.
- Focus on improving the perception of drones by making them robust to noise, sensor failures and adversarial interference.
- Major contribution involves integrating heterogeneous sensor streams from camera and LiDAR.

Independent Researcher / Freelance Consultant

Self-Employed, Salem, India

Jul 2024 - Apr 2025

- Conducted independent research in Explainable AI and Online Learning, contributing to academic discussions on AI.
- Authored a book chapter titled "Bridging Online Learning and Explainability", published in the book "Explainable AI for Earth Observation Data Analysis: Challenges, Opportunities and Applications"

Student Intern

ABB Corporate Research Center, Bengaluru, India

Jul 2023 - Jul 2024

- Conducted research on collaborative learning techniques for model adaptation under distribution drift scenarios.
- Developed an algorithm for efficient drift detection and target model adaptation with reduced communication and computational costs.
- Developed an internal simulation tool for federated learning experiments.

TEAM WORK EXPERIENCE

Data Analyst Intern

Jun 2021 - Jul 2022

Urjanet Energy Solutions Pvt Ltd, Chennai, India

- Validated user-centric scenarios by extracting, loading, and reconciling large amounts of data from various platforms and sources.
- Maintained utility data for multinational companies, including analyzing and storing data.
- Addressed various incident reports by drafting and providing solutions to developers.
- Resolved customer inquiries directly and ensured customer satisfaction.

Web Developer Intern

May 2020 - Jul 2020

Xtracut Systems Pvt Ltd, Chennai, India

- Managed and maintained eCommerce websites for clients.
- Collaborated with project managers, testers, and end-users to ensure successful project

- completion.
- Utilized scripting languages, content creation tools, and digital media to design and develop websites.
- Used management tools like Jira to track and organize projects.

EDUCATION

University of Twente, The Netherlands

May 2025 - Apr 2029

Doctor of Philosophy

Area of Research: Computer Vision, Mutli-modal AI, Autonomous Systems, Multi-Robot Systems

Dissertation: Multi-Platform Multi-modal Information Fusion in Dynamic Environments for Drone Swarms (Expected: Apr 2029)

Indian Institute of Information Technology, Sri City, India

Sep 2022 - Jun 2024

Master of Technology - Artificial Intelligence & Machine Learning - *CGPA - 9.55/10*

Area of Research: Federated Learning, Concept drift, Distributed systems, Autonomous Systems

Dissertation: Collaborative Drift Compensation

Anna University, Easwari Engineering College, Chennai, India

Sep 2018 - Apr 2022

Bachelor of Technology - Information Technology - *CGPA - 8.64/10*

SERVICE

Feb 2025 Reviewer IEEE Internet Computing.

Feb 2025 Reviewer TrustNLP 2025.

Jan 2025 Reviewer Knowledge-Based Systems.

Sep 2024 Reviewer NeurIPS Bayesian Decision-making and Uncertainty Workshop 2024.

Aug 2024 Reviewer AlphaXiv - Stanford University project.

PUBLICATIONS

A. N. Latha, S. N. Anbalagan, A. Chiumento, and J. Laarhuis. "Where Do We Go from Here? Charting the Future of Unmanned Vehicles." In *Proc. 2025 16th IFIP Wireless and Mobile Networking Conference (WMNC)*, Leuven, Belgium, 2025, pp. 176–181, doi: 10.23919/WMNC67099.2025.11299247.

Adarsh N. L., P. V. Arun, and B. K. Mohan. "Bridging Online Learning and Explainability in Image Classification." In *Explainable AI for Earth Observation Data Analysis: Applications, Opportunities, and Challenges*, edited by P. V. Arun, J. Chanussot, B. K. Mohan, D. N. Kumar, and A. Porwal, pp. 191–203. CRC Press, 2025. doi: 10.1201/9781003597025-11.

Adarsh N. L., Madapu Amarlingam, and Divyasheel Sharma. "Collaborative Drift Compensation." In *Proc. 6th Joint International Conference on Data Science and Management of Data (CODS-COMAD 2024)*, 2024. doi: 10.1145/3703323.3703341.

Adarsh N. L. and P. V. Arun. "Leveraging Internal Representations of Model for Magnetic Image Classification." In *Proc. Computer Vision and Image Processing (CVIP) 2024*, Springer, 2024.

Madapu Amarlingam, Abhishek Wani, and **Adarsh N. L.** "Lightweight Industrial Cohorted Federated Learning for Heterogeneous Assets." Submitted to *Knowledge-Based Systems*, 2024. (Also available as arXiv:2407.17999).

Adarsh N. L., P. V. Arun, and Aravindh N. L. "Enhancing Image Caption Generation Using Reinforcement Learning with Human Feedback." arXiv preprint arXiv:2403.06735, 2024.

Ananthi N., **Adarsh N. L.**, Anish M., and Arunraj G. "Person Re-Identification using Deep Learning." In *Proc. 2023 7th International Conference on Computing Methodologies and Communication (ICCMC)*, 2023, pp. 127–132, doi: 10.1109/ICCMC56507.2023.10083656.

R. Hemalatha, S. Nirmala, K. Viswaja, **N. Adarsh**, Annie Catherine, and Priyatharshini Rajaram. "Evaluation of the Effectiveness of DAD-APP in Managing Dentally Anxious Children: A Pilot Study." *Zeichen Journal*, vol. 6, pp. 119–126, 2020.